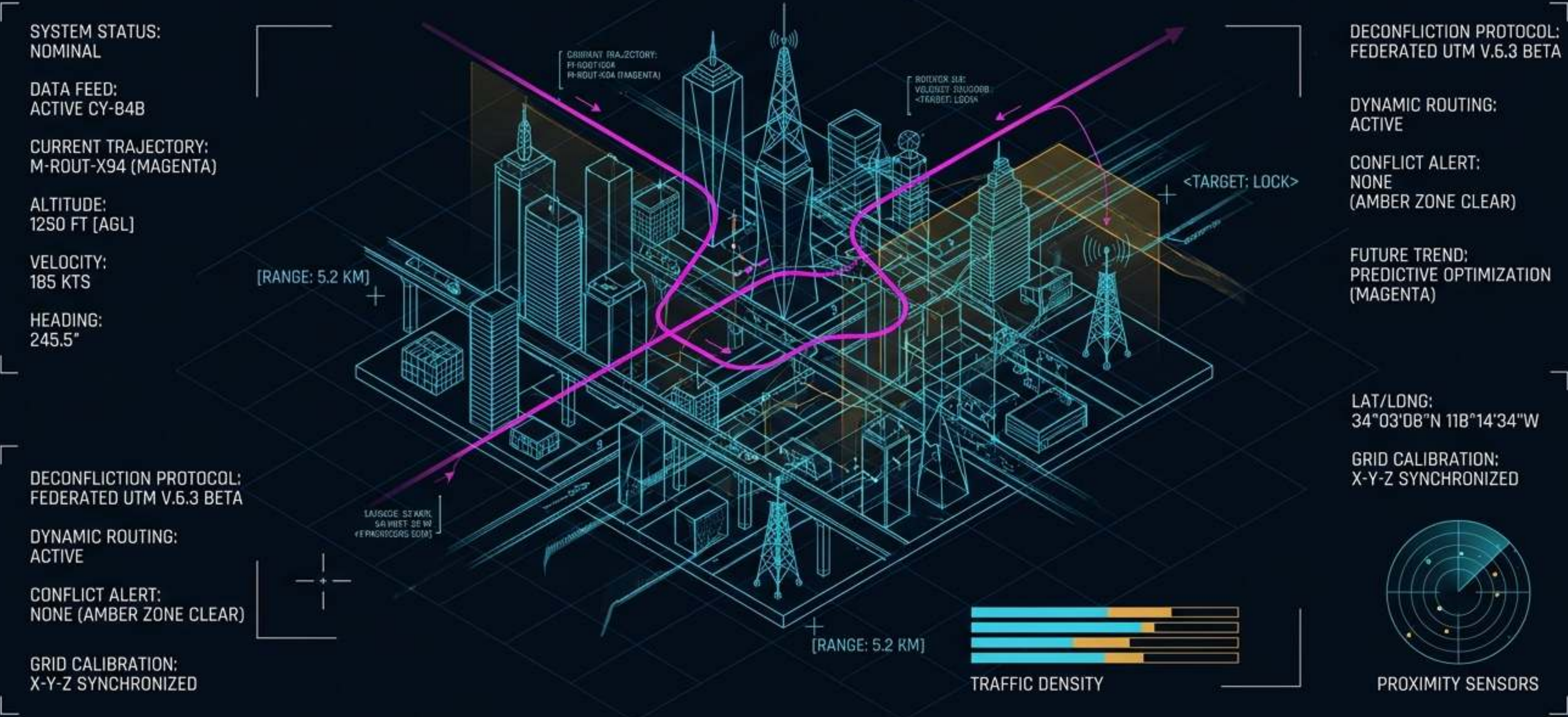


THE BLUEPRINT OF AUTONOMOUS AIRSPACE

6.3 UTM, ALGORITHMIC DECONFLICTION, AND THE ARCHITECTURE OF FEDERATED AIRSPACE MANAGEMENT.

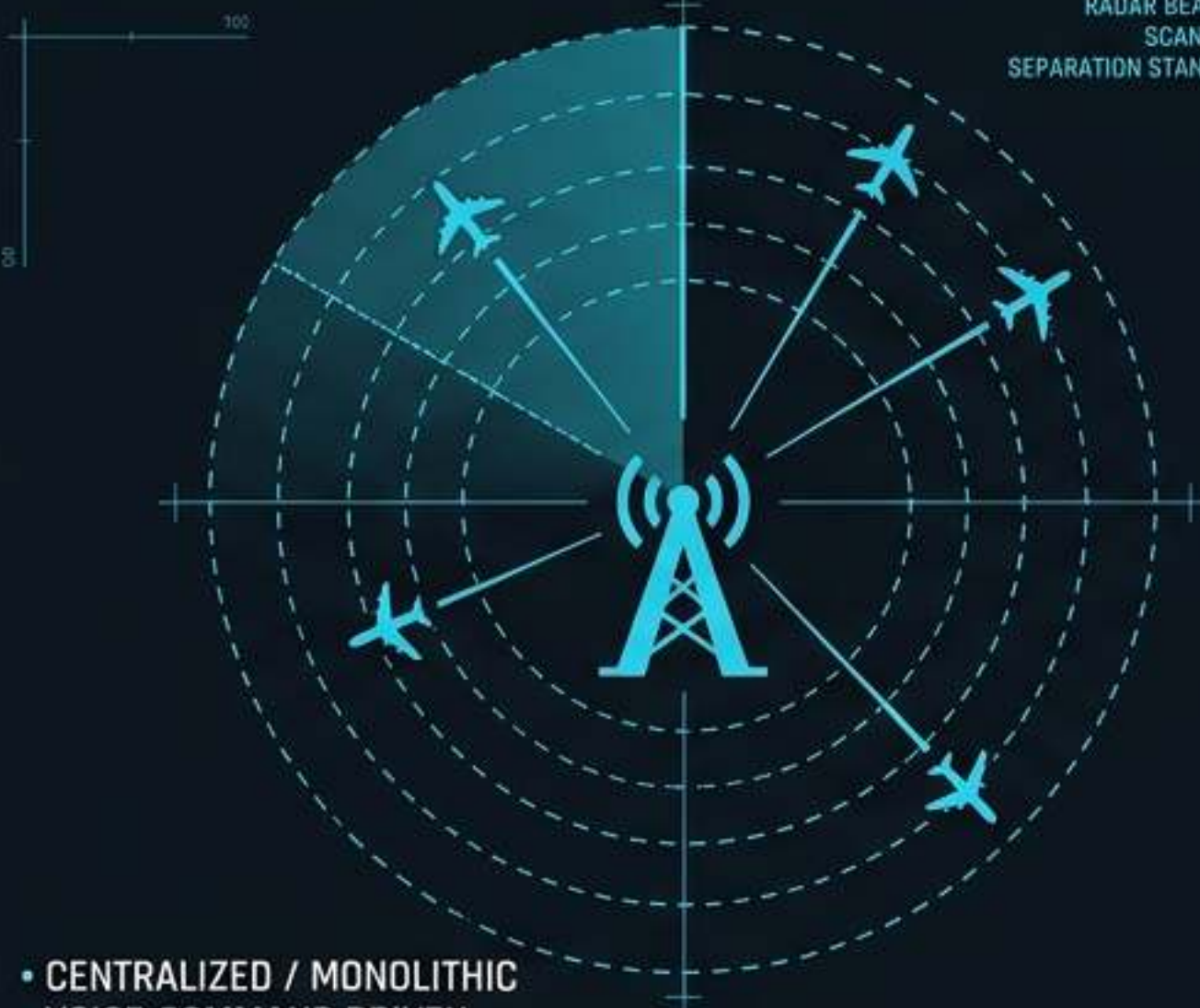


SCALING THE AIRSPACE: FROM CENTRALIZED CONTROL TO FEDERATED NETWORKS

GLOBAL AIRSPACE LOAD: 85%
AUTOMATION EFFICIENCY: +40%
INCIDENT RATE: < 0.00001%

LEGACY ATM

RADAR BEAM WIDTH: 1.5°
SCAN RATE: 15 RPM
SEPARATION STANDARD: 3-5 NM



- CENTRALIZED / MONOLITHIC
- VOICE-COMMAND DRIVEN
- HUMAN-IN-THE-LOOP RADAR SEPARATION

TRANSITION
GATEWAY

DIGITAL
INTEGRATION
BRIDGE

UTM PARADIGM

NETWORK LATENCY: < 10 ms
DISTRIBUTED TRUST NODES: N=64
AUTOMATION LEVEL: L4/L5



- FEDERATED / DISTRIBUTED
- DIGITAL INTENT / API-DRIVEN
- AUTOMATED MANAGE-BY-EXCEPTION

MILESTONE TRACKING: 2024-Q3

SYSTEM READINESS LEVEL: 6

GLOBAL AIRSPACE LOAD: 85%
AUTOMATION EFFICIENCY: +40%
INCIDENT RATE: < 0.00001%

BUDGET ACTIVITY 6.3 (RDT&E) CORE MANDATE: ENSURE INTEROPERABILITY, ALGORITHMIC FAIRNESS, AND ZERO DEGRADATION OF EXISTING ATM SAFETY MARGINS WHILE TRANSITIONING FROM APPLIED RESEARCH TO OPERATIONAL PROTOTYPES.

SYSTEM READINESS LEVEL: 6

SAFETY ASSURANCE CASE: IN PROGRESS

U-SPACE VOLUME CLASSIFICATIONS AND SEPARATION PROVISIONS

GLOBAL AIRSPACE LOAD: 70%
AUTOMATION EFFICIENCY: +45%
INCIDENT RATE: < 0.00001%



VOLUME	TRACKING	SEPARATION	RESOLUTION
VOLUME X (UNCONTROLLED)	Tracking: Not mandated	Visual / Remain Well Clear	No tactical conflict resolution
VOLUME Y (UNCONTROLLED, PREFLIGHT SHARING)	Tracking: Mandatory	Bubble-based pre-flight deconfliction	Strategic Conflict Resolution provided
VOLUME ZU (U-SPACE CONTROLLED)	Tracking: Mandatory	Live tracking and automated alerts	Tactical Conflict Resolution provided by U-space (ground-computed)
VOLUME ZA (ATM CONTROLLED)	Tracking: Mandatory	Standard ATM minimums	Tactical separation provided by Air Traffic Control (human-in-the-loop)

GLOBAL AIRSPACE LOAD: 70%
AUTOMATION EFFICIENCY: +45%
INCIDENT RATE: < 0.00001%

MILESTONE TRACKING: 2024-03

SYSTEM READINESS LEVEL: 6

INSIGHT: THE SHIFT FROM Y TO ZU REPRESENTS THE TRANSITION FROM STRATEGIC DECONFLICTION (PRE-FLIGHT TRAJECTORY PLANNING) TO TACTICAL DECONFLICTION (REAL-TIME AUTOMATED ROUTING ADJUSTMENTS).

SYSTEM READINESS LEVEL: 6

SAFETY ASSURANCE RAGE: IN PROGRESS

THE ALGORITHMS OF SAFETY: ICAO THREE-LAYER CONFLICT MANAGEMENT

SYSTEM SAFETY RELIES ON SEQUENTIAL ALGORITHMIC DEGRADATION: UTM NETWORKS HANDLE STRATEGIC AND TACTICAL HORIZONS, WHILE ONBOARD AVIONICS PROVIDE THE FINAL FAILSAFE.

LAYER 1: STRATEGIC DECONFLICTION (PRE-FLIGHT)

- Governed by ASTM F3548-21.
- Focuses on arrangement and prioritization of 4D Operational Intents (OI).
- Objective: Minimize conflict likelihood before takeoff.

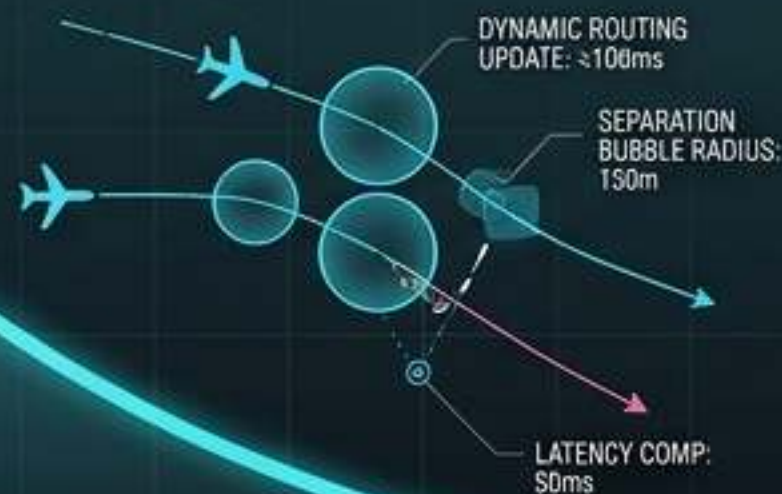
PRE-FLIGHT
OPTIMIZATION

OI PRIORITIZATION

CONFLICT PROBABILITY
REDUCTION: <1%

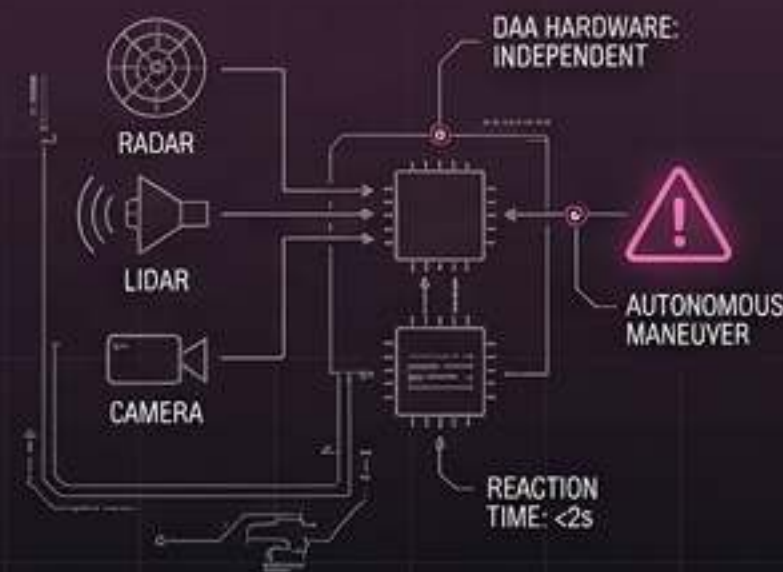
LAYER 2: TACTICAL DECONFLICTION (IN-FLIGHT)

- Focuses on separation provision.
- Employs dynamic routing and bubble-based separation parameters.
- Tailored to aircraft speed, telemetry latency, and weather.



LAYER 3: COLLISION AVOIDANCE (DAA)

- The last line of defense.
- Exclusively onboard Detect and Avoid hardware.
- Independent of the UTM network.



HOURS

MINUTES

SECONDS

0

10

30

40

60

TIME TO INCIDENT (HOURS → MINUTES → SECONDS)

MATHEMATICAL AIRSPACE: DEFINING OPERATIONAL INTENT (OI)

SPATIAL CONSTRAINTS (3D)

Precise geospatial coordinates (Latitude, Longitude, Altitude) creating a volumetric safety buffer.

TEMPORAL CONSTRAINTS (1D)

Specific entry and exit timestamps governing the occupation of the 4D volume.

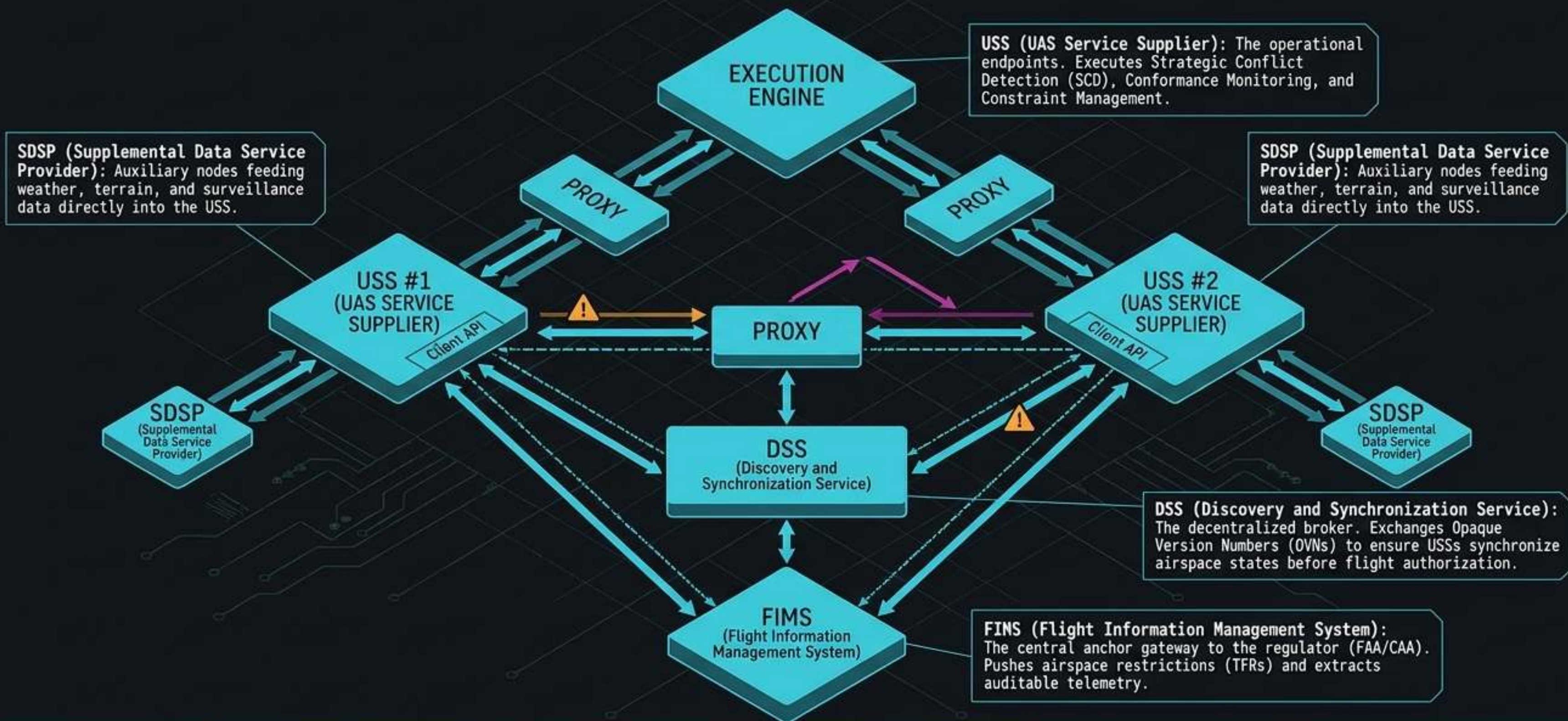
ASTM F3548-21 STANDARD

Dictates the USS Interoperability required to detect and resolve intersecting 4D volumes before departure.

STRATEGIC CONFLICT DETECTION (SCD)

Continuous network querying to ensure a newly proposed OI is free of 4D intersection with all known operations.

THE UTM TECH STACK: FEDERATED NETWORK ARCHITECTURE



PROTOCOL STANDARDIZATION: REMOTE IDENTIFICATION ARCHITECTURE

NETWORK RID (Net-RID)

MECHANISM:

Telemetry sent via cellular/Wi-Fi to a USS; published via DSS.

RANGE:

Global (Internet-dependent).

SECURITY:

High (Asymmetric cryptography, HHIT).

BROADCAST RID (B-RID)

MECHANISM:

Direct RF-LOS broadcast (Bluetooth/Wi-Fi) from the UA.

RANGE:

Local (Direct line-of-sight).

SECURITY:

Open Drone ID formats; vulnerable to spoofing without TBRD (Tesla Authenticated Broadcast).

CROWD-SOURCED RID (CS-RID / DRIP)

MECHANISM:

Gateways bridge B-RID into UTM via smartphone receivers.

RANGE:

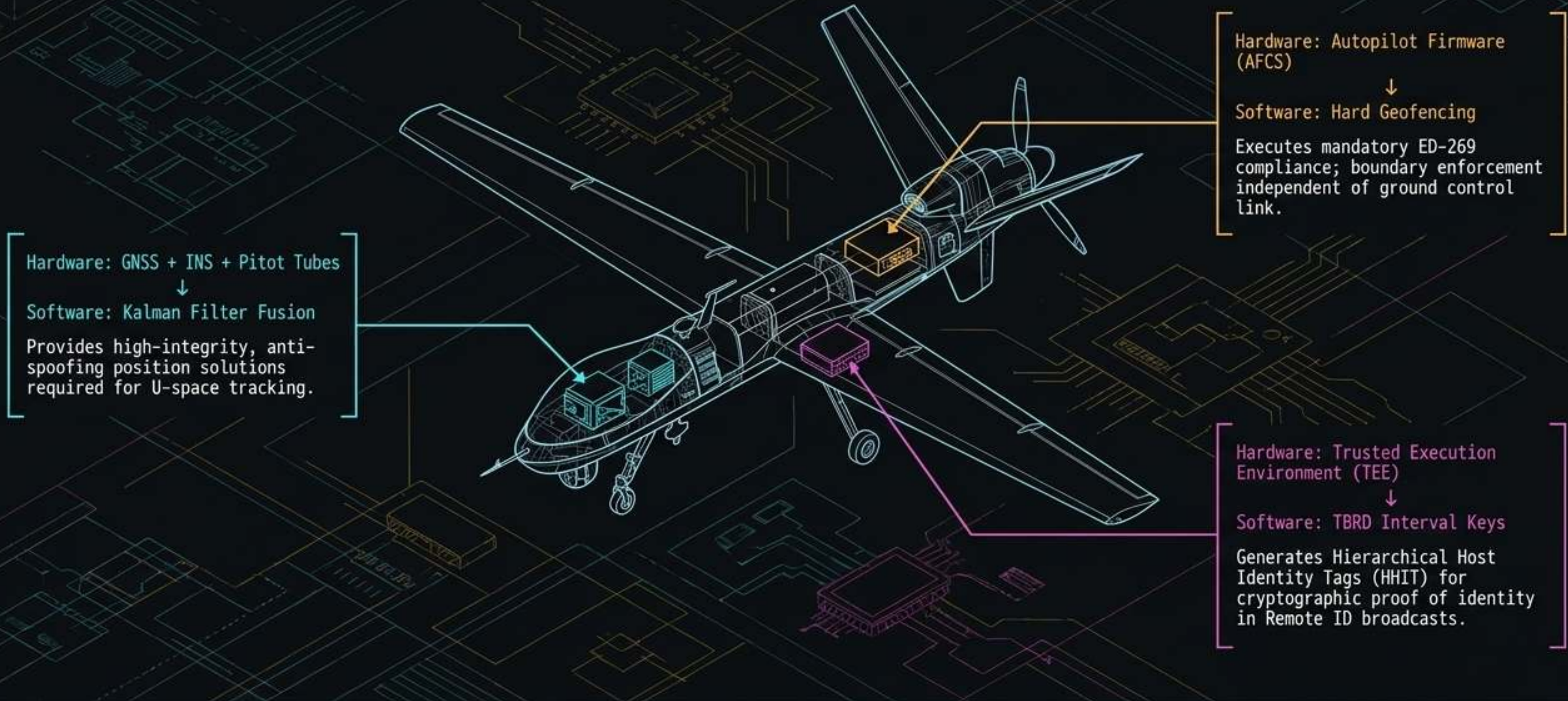
Expanded local; uses Multilateration (Time of Arrival) to verify physical UA position.

SECURITY:

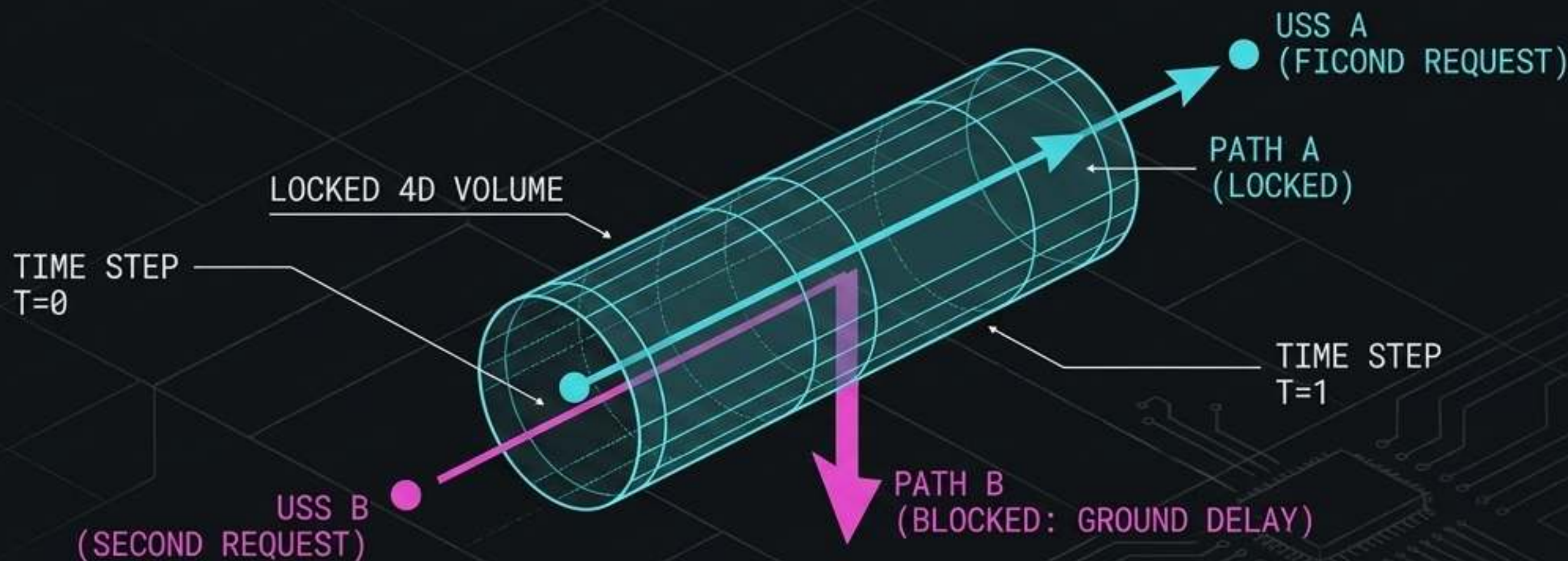
Defeats intentional deception via independent physical layer verification.

Dynamic routing →

INTEGRATION X-RAY: LINKING AVIONICS TO UTM LOGIC



SYSTEM UNDER STRESS: THE FIRST-REQUESTED-FIRST-SERVED (FRFS) PROBLEM



THE MECHANISM

UTM relies on FRFS. The first operator to file an Operational Intent (OI) locks the 4D volume. The second operator must absorb the ground delay to deconflict.

THE INEQUITY

Without central flow control, operators who file further in advance gain massive competitive advantages, forcing competitors to bear the entire cost of deconfliction delay.

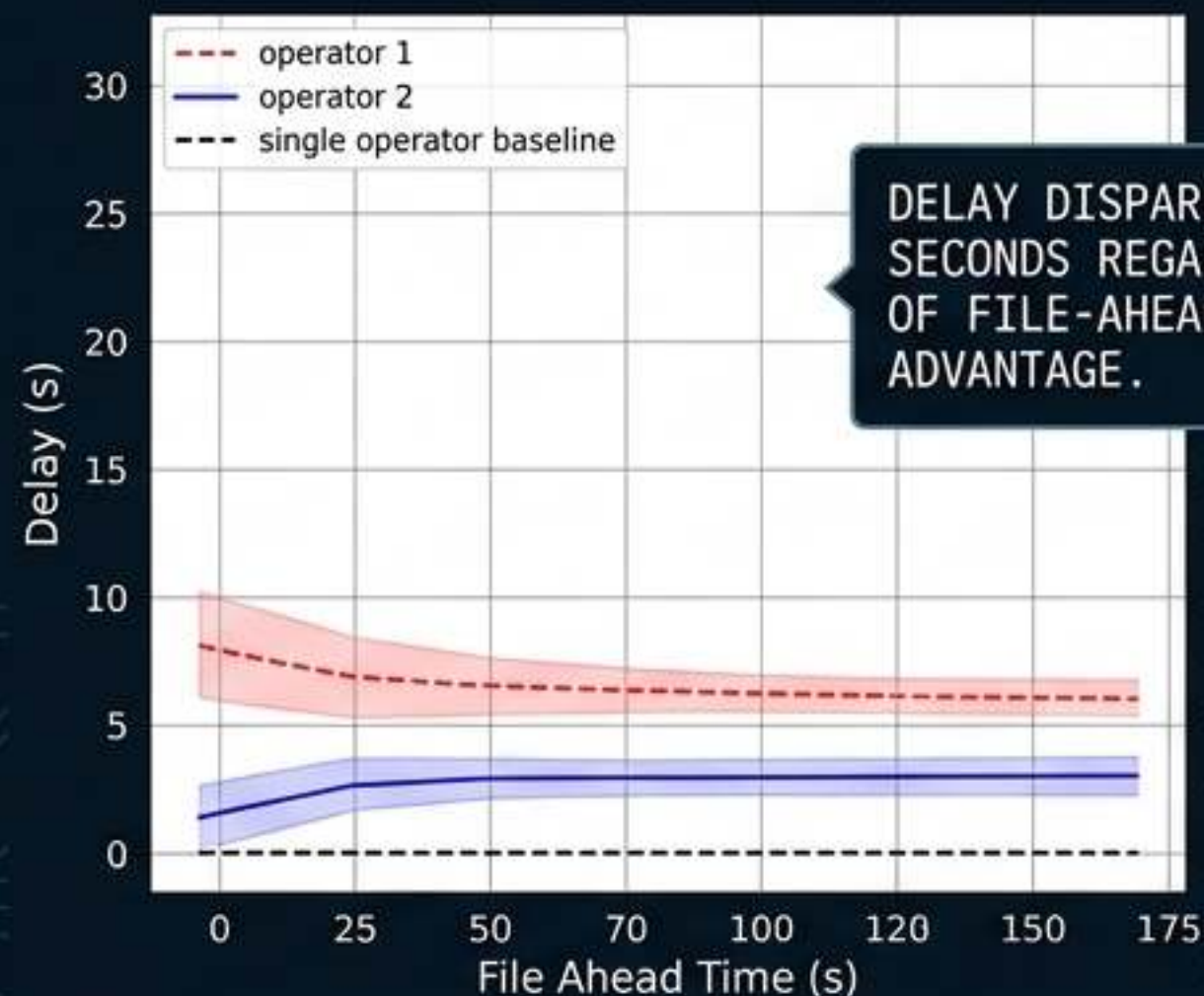
THE REQUIREMENT

A mathematical definition of fairness. Measured by the normalized ratio of geometric and arithmetic mean delays across operators.

[FAIRNESS METRIC: $(G_{\text{mean}} / A_{\text{mean}})$]

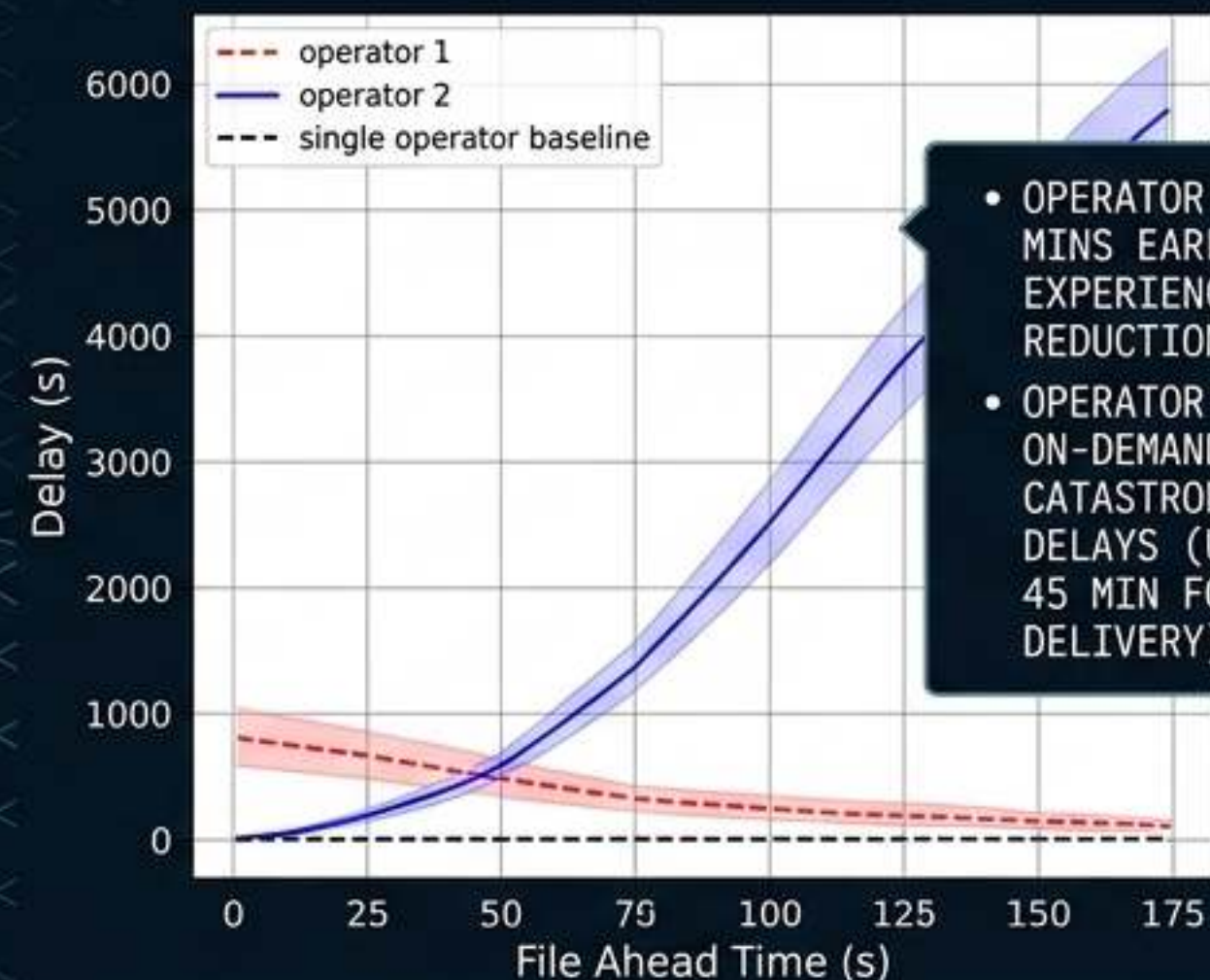
ALGORITHMIC INEQUITY AT SCALE: AIRBUS/NASA SIMULATION DATA

LOW DENSITY (25 REQ/HR)



DELAY DISPARITY IS <10 SECONDS REGARDLESS OF FILE-AHEAD ADVANTAGE.

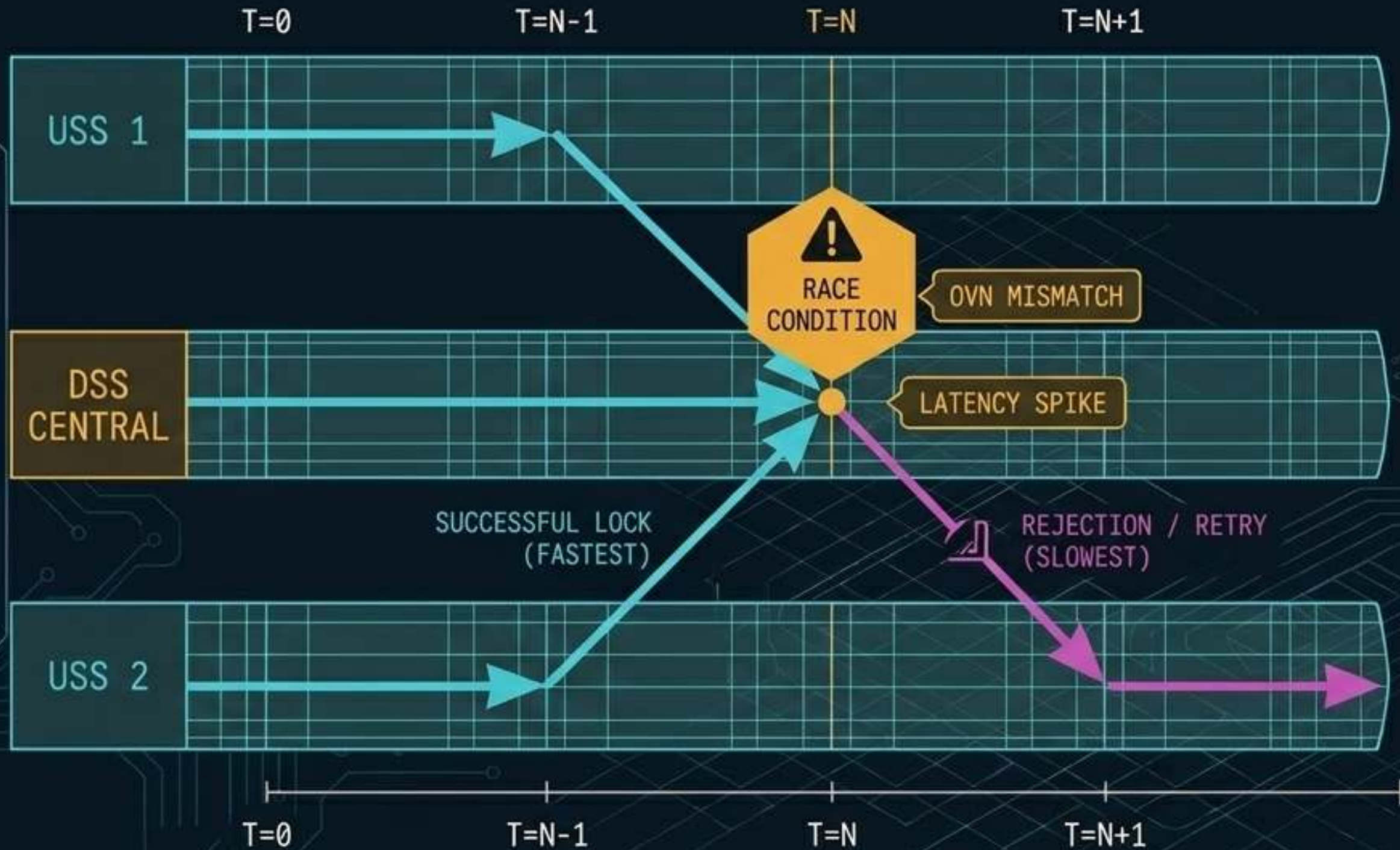
HIGH DENSITY (250 REQ/HR)



- OPERATOR 1 (FILES 30 MINS EARLY): EXPERIENCES A 70% REDUCTION IN DELAY.
- OPERATOR 2 (FILES ON-DEMAND): ABSORBS CATASTROPHIC GROUND DELAYS (UP TO 1 HR 45 MIN FOR PACKAGE DELIVERY).

THE ENGINEERING SOLUTION: IMPLEMENTING A REQUIRED TIME TO ACT (RTTA)—A FREEZE HORIZON RESTRICTING HOW EARLY AN OI CAN BE LOCKED, RESTORING THE NORMALIZED FAIRNESS METRIC TO ≥ 0.95 .

NETWORK VULNERABILITIES: RACE CONDITIONS AND DSS LATENCY



TECHNICAL INSIGHTS

- **THE SYNCHRONIZATION CHALLENGE:** THE DISCOVERY AND SYNCHRONIZATION SERVICE (DSS) RELIES ON OPAQUE VERSION NUMBERS (OVNS) TO GUARANTEE USSs ARE USING CURRENT AIRSPACE STATES.
- **THE RACE CONDITION:** AT HIGH TRAFFIC SCALES, TWO USSs MAY SIMULTANEOUSLY RUN STRATEGIC CONFLICT DETECTION (SCD) ON THE SAME OBSOLETE OVN.
- **ANAMLL SIMULATION FINDINGS:** MIT LINCOLN LABORATORY TESTS REVEAL THAT TIME-CORRELATED DEMAND SPIKES CAUSE HIGH REJECTION RATES. CONNECTIVITY LATENCY CREATES AN ASYMMETRIC COMPETITIVE ADVANTAGE FOR FASTER USS NODES.
- **ARCHITECTURAL FIX:** REQUIRES REFINED PRIORITIZATION PROTOCOLS AND ROBUST RETRY-LOGIC WITHIN THE USS APIS TO HANDLE OVN MISMATCHES SMOOTHLY.

PROVING THE ECOSYSTEM: NASA TCL 3 & 4 OPERATIONS

DYNAMIC UVRs (UNMANNED VIRTUAL ROUTING)

The system successfully established temporary restricted areas in real-time, instantly pushing automated alert messages to affected operators to execute return-to-launch procedures.

TEMPORARY
RESTRICTED
AREA (UVR)

! REROUTING ALERT

SIM PACKAGE

SIM S&R

LIVE/SIM INTEGRATION

- Seamlessly managed airspace containing both physical drones and simulated traffic (package delivery, search & rescue) to safely artificially stress the USS network.

URBAN COMPLEXITY MANAGEMENT

- Proven capability to execute BVLOS operations amidst wind shears, high obstacle density, and mixed radio/cellular communication environments.

NEXT-GEN RESILIENCE: 6G & QUANTUM DEFENSE



6G TELECOMMUNICATIONS

Shifting beyond 5G/LTE. Required for ultra-low latency, highly reliable BVLOS command and control (C2) links, enabling true real-time tactical deconfliction in dense urban canyons.



QUANTUM-RESISTANT CRYPTOGRAPHY

Upgrading the asymmetric cryptography used in HHIT and DRIP protocols to protect UTM data integrity against future quantum computing decryption capabilities.



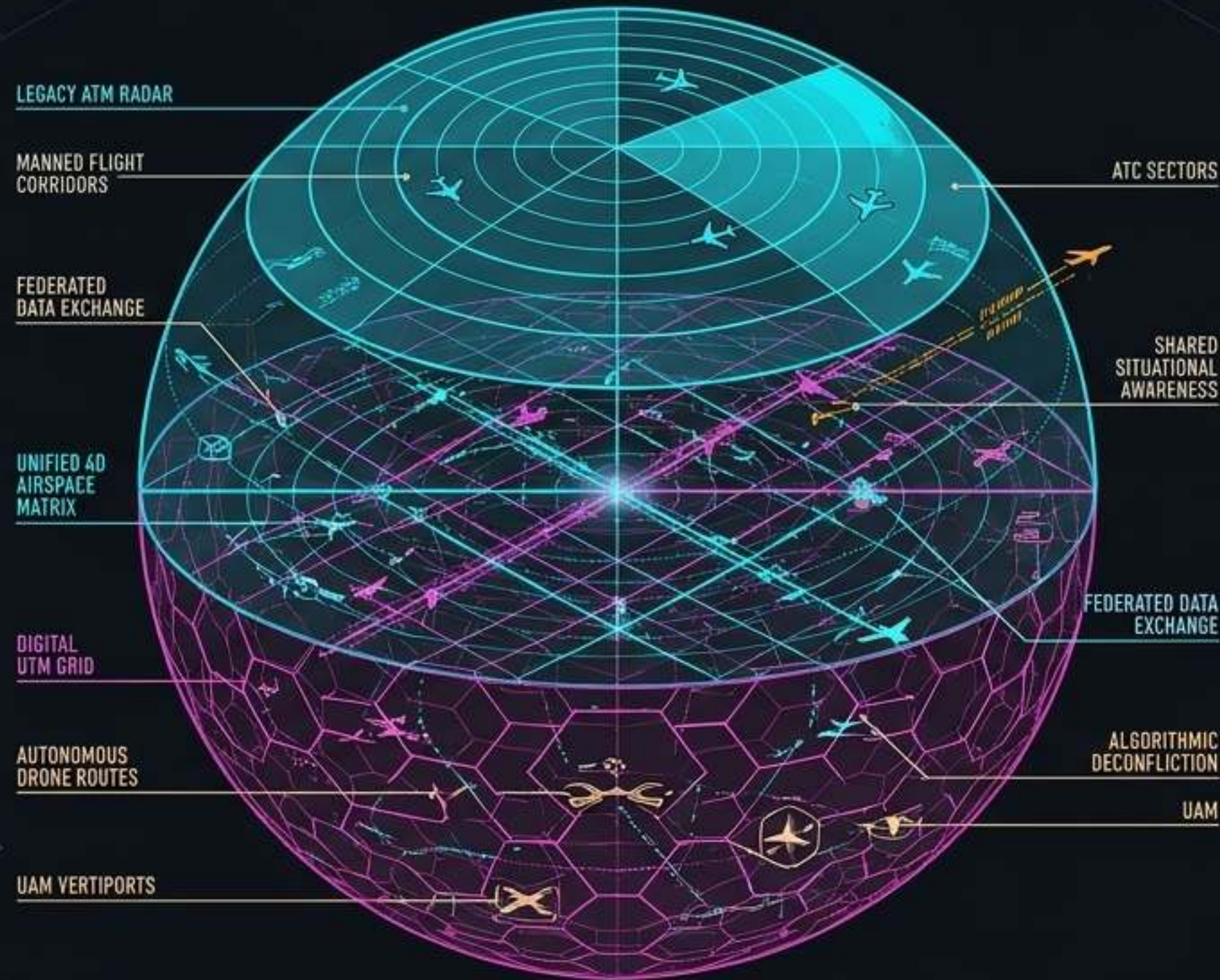
DECENTRALIZED LEDGERS (ORBIT DB)

Transitioning telemetry logging to blockchain-based architectures to ensure flight records are immutable, auditable, and distributed securely across the federated network.

THE BLUEPRINT REALIZED: FULL ATM/UTM CONVERGENCE

SYNTHESIS INSIGHT

Budget Activity 6.3 is not building a separate sky; it is prototyping the foundational software core for all future low-altitude air traffic.



THE CONVERGENCE

By proving federated architecture, enforcing ASTM standards, and solving algorithmic fairness, UTM establishes the framework where automated drones, UAM air-taxis, and legacy manned aviation share a single, digitally negotiated 4D airspace.

FROM HUMAN-IN-THE-LOOP TO ALGORITHMIC CERTAINTY: THE FUTURE OF AVIATION IS FEDERATED, AUTONOMOUS, AND SEAMLESSLY INTEGRATED